

Backpropagation, Part 3: Why the Update Works

Contents

1. Overview	2
2. Prerequisites	3
3. The loss is a function of all nine parameters	3
4. What the gradient is	5
5. What a derivative tells us	5
6. Minimum by setting the derivatives to zero	6
7. Why we subtract the derivative	7
7.1 One parameter at a time	7
7.2 The sign decides the direction	8
8. Why we need a learning rate	9
9. When to stop: convergence	10
10. Summary	12
11. Sources	12
12. Key terms	12

Where this fits: Pipeline map steps 8 (Model), 10 (Tune). See the [Course map](#).



- **Builds on:** Feature scaling ([Note 24](#)); Best-fit line and squared error ([Note 51](#)); Sigmoid function ([Note 74](#)); Derivatives of one variable ([Note 600](#)); Partial derivatives and gradients ([Note 601](#)); Hessian and multivariate Taylor ([Note 603](#)).
- **Leads to:** Vanishing gradient ([Note 1018](#)); Batch gradient descent ([Note 1020](#)); Stochastic gradient descent ([Note 1020](#)); Mini-batch gradient descent ([Note 1020](#)); Improving a neural network ([Note 1021](#)); Scaling inputs for neural networks ([Note 1023](#)).
- **Compare with:** Ordinary least squares (closed form) ([Note 51](#)); Normal equation ([Note 55](#)).

1. Overview

Key point: The loss of a network is one function of all its weights and biases. Each derivative tells how the loss responds to one parameter, in size and sign. Stepping against it, by a learning rate times its size, walks every parameter downhill until the slope reaches zero.

The [backpropagation what Note](#) and the [backpropagation how Note](#) showed what backpropagation does and how to code it. All the learning happens in one line, the update

$$W_{\text{new}} = W_{\text{old}} - \eta \frac{\partial L}{\partial W}$$

This Note explains why that line works. The Note reuses ideas from earlier Notes (derivatives, gradients, minima, the learning rate) and applies them to the 2-2-1 regression network.

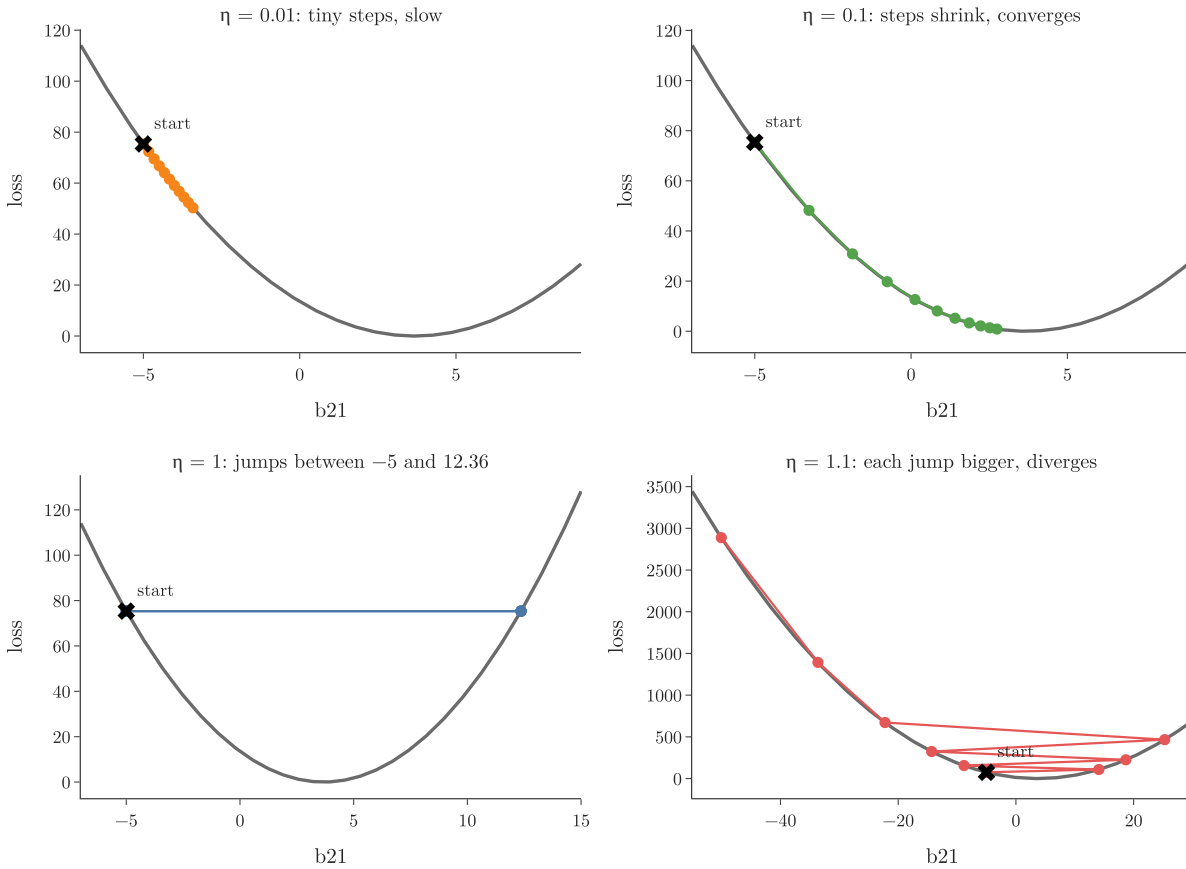


Figure 1: The loss as a function of the output bias b_{21} alone, and 10 updates of b_{21} from -5 with four learning rates

2. Prerequisites

- The [backpropagation](#) [what Note](#): the 2-2-1 network, the four students and the 9 derivatives.
- [Gradient descent](#), including its learning rate and stopping rule.
- [Derivatives](#) and [partial derivatives and gradients](#).

3. The loss is a function of all nine parameters

Key point: Write $\hat{y}$ out in full and the inputs and y become constants: what is left is a function of the 6 weights and 3 biases. Training is turning 9 knobs until the loss is smallest.

For the regression network, $L = (y - \hat{y})^2$. The package y comes from the data, so for a given student it is a constant. Only $\hat{y}$ can vary, and $\hat{y}$ is built from the hidden outputs:

$$\hat{y} = W_{11}^2 O_{11} + W_{21}^2 O_{12} + b_{21}$$

Putting in what O_{11} and O_{12} are writes the whole network as one formula:

$$\hat{y} = W_{11}^2 (W_{11}^1 x_{i1} + W_{21}^1 x_{i2} + b_{11}) + W_{21}^2 (W_{12}^1 x_{i1} + W_{22}^1 x_{i2} + b_{12}) + b_{21}$$

In this formula x_{i1} and x_{i2} are the two **features** of student i (input variables, one column of the data table each: CGPA and profile score), so they are constants. Everything else is a parameter:

$$L = L(W_{11}^1, W_{12}^1, W_{21}^1, W_{22}^1, b_{11}, b_{12}, W_{11}^2, W_{21}^2, b_{21})$$

$y = f(x)$ is a function of one variable; the loss is a function of nine. These nine numbers are the network's **trainable parameters** (G-1999): the values training is allowed to change. Picture the network as a box with 9 knobs, one per trainable parameter: turning any of them changes the loss, and training turns all of them until the loss is as small as possible.

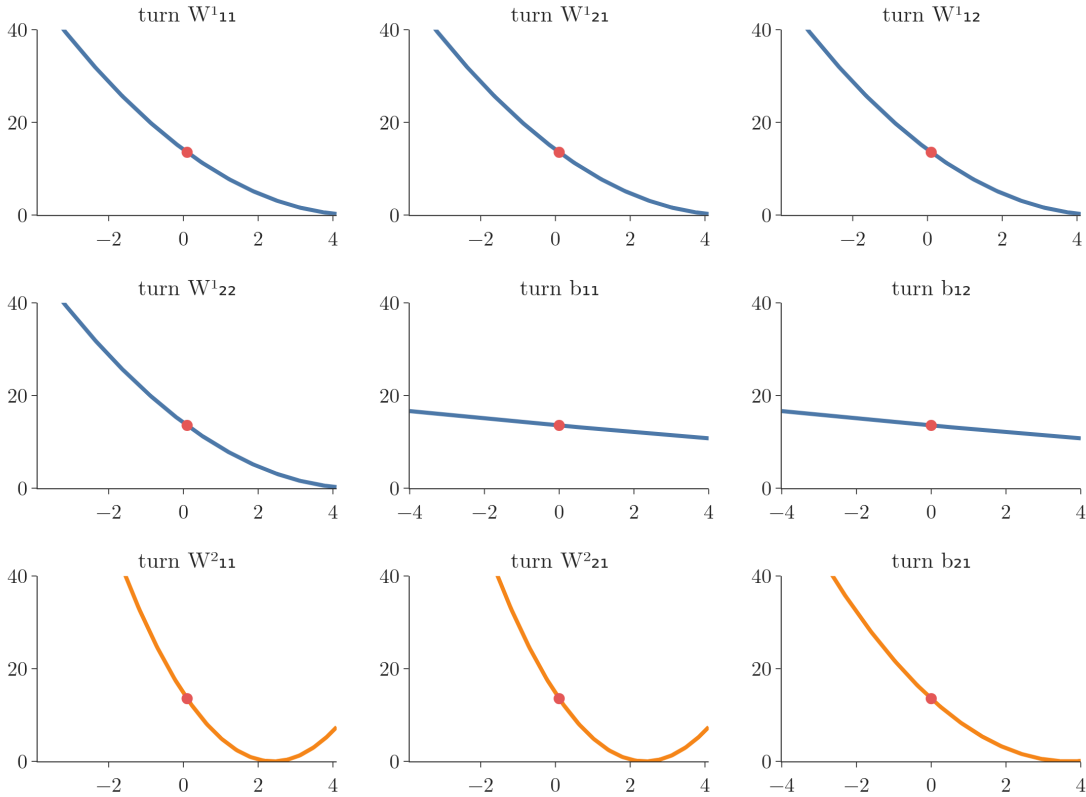


Figure 2: Student 1's loss when one parameter is turned and the other eight stay at their starting values; the red dot is the start

Figure 2 turns each knob on its own. Every curve passes through the same red dot, the starting loss 13.54, but each falls at its own rate. The hidden biases b_{11} and b_{12} barely move it: their slope at the start is only -0.736 , against -5.888 for the first-layer weights and -11.776 for the output weights (the gradients of the [backpropagation what Note](#), section 7.1). The knob picture is the general idea of section 2 of the [convex and non-convex cost functions Note](#), for this network.

4. What the gradient is

Key point: One derivative per parameter, each with the others held fixed. For our loss that is 9 partial derivatives; together they are the gradient.

For a function of one variable, such as $y = x^2 + x$, the derivative is $dy/dx = 2x + 1$. For a function of several variables, such as $z = x^2 + y^2$, we take one **partial derivative** per variable, holding the others fixed: $\partial z/\partial x = 2x$ and $\partial z/\partial y = 2y$. The **gradient** collects them all (see the [partial derivatives and gradients Note](#)).

So “computing the gradient of the loss” means computing its 9 partial derivatives, one per knob: exactly the 9 formulas of the [backpropagation what Note](#). Geometrically, each one is the slope of the loss along one of 9 directions.

Extra: A gradient is more than “a fancy word for a derivative”. The gradient is the vector of all partial derivatives, and as an arrow it points in the direction in which the loss rises fastest (section 4.2 of the [partial derivatives and gradients Note](#)). Gradient descent steps along the opposite arrow, which is where its name comes from.

5. What a derivative tells us

Key point: A derivative is a rate of change. Its size says how strongly the loss reacts, and its sign says in which direction.

A derivative is a **rate of change**: $dy/dx = 2$ means that increasing x by a small amount increases y by twice that amount; $dy/dx = -2$ means y decreases by twice that amount. Both the size and the sign carry information (see the [derivatives Note](#)).

A derivative at a point is the slope there. For $y = x^2 + x$, $dy/dx = 2x + 1$; at $x = 5$ it is 11, so near $x = 5$ the function rises 11 times as fast as x .

For the network, $\partial L/\partial W_{11}^1$ says how the loss responds to a small change in W_{11}^1 . For student 1 at the start it is -5.888 : raising W_{11}^1 by 0.001 lowers the loss by about 0.006.

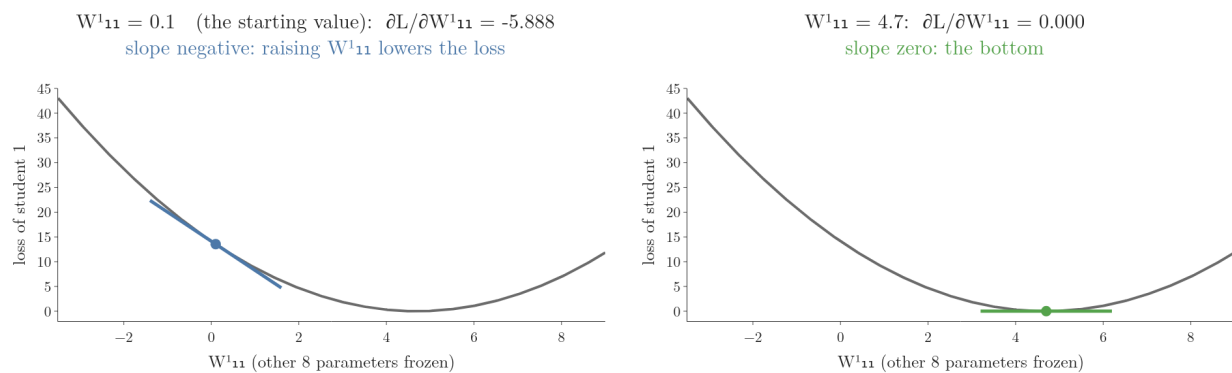


Figure 3: The **tangent line** (G-1945) sliding along student 1’s loss as a function of W^1_{11} ; its slope is the derivative

Figure 3 slides the tangent along the loss curve of W^1_{11} . Watch the slope’s sign: negative (blue) left of the bottom, zero at $W^1_{11} = 4.7$, positive (orange) to the right; and its size: steep far from the bottom, flat near it.

6. Minimum by setting the derivatives to zero

Key point: At a minimum every slope is zero. For $z = x^2 + y^2$ we can solve that by hand; for a network’s loss we cannot, so we walk downhill instead.

At the lowest point of a smooth curve the slope is zero. For $y = x^2$, $dy/dx = 2x = 0$ gives $x = 0$. With two variables, $z = x^2 + y^2$, both partial derivatives must be zero: $2x = 0$ and $2y = 0$, so the minimum is at $(0, 0)$ (Figure 4).

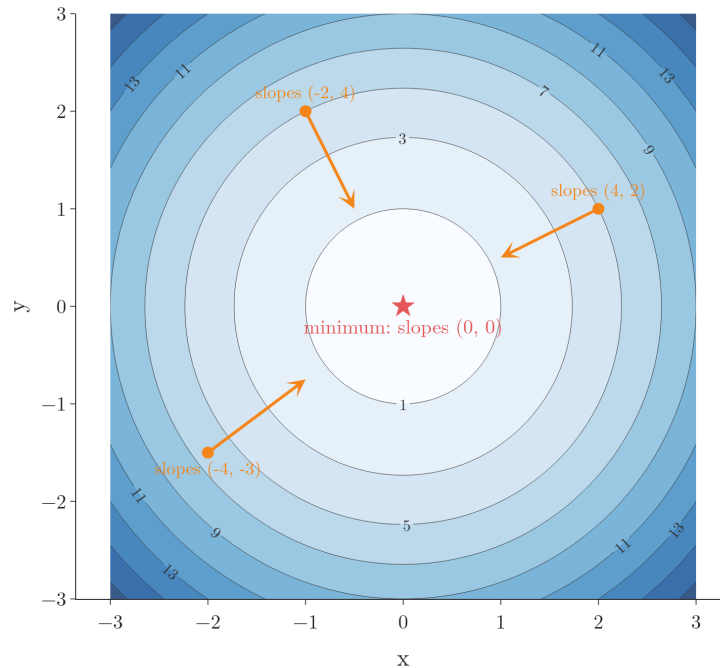


Figure 4: Contours of $z = x^2 + y^2$. At three points the arrow is minus the two slopes; only at the star are both slopes zero

In Figure 4, every arrow, minus the pair of slopes, points towards the centre; at the centre both slopes are zero and there is nowhere lower to go.

For our network the same idea says: set all 9 partial derivatives to zero and solve. But those 9 equations are tangled together through products such as $W_{11}^2 W_{11}^1$, and for any real network there are thousands of them, with no formula for the solution: networks are trained with iterative, gradient-based methods instead (Goodfellow et al. 2016, §6.2). So instead of solving, we start somewhere and walk downhill with gradient descent, as the [gradient descent Note](#) does for linear regression.

7. Why we subtract the derivative

Key point: A positive slope means the loss rises as the parameter rises, so we lower the parameter; a negative slope means the opposite. The minus sign handles both cases in one rule.

7.1 One parameter at a time

Key point: Freeze the other 8 parameters: for student 1 the loss in b_{21} alone is the parabola $(3.68 - b_{21})^2$, lowest at $b_{21} = 3.68$.

To see the update in action, freeze 8 of the 9 parameters at their starting values (weights 0.1, other biases 0) and keep only b_{21} free. For student 1 the forward pass gives $0.32 + b_{21}$, so

$$L(b_{21}) = \left(4 - (0.32 + b_{21})\right)^2 = (3.68 - b_{21})^2$$

The loss $L(b_{21})$ is a parabola with its lowest point at $b_{21} = 3.68$ (Figure 1). Its slope is

$$\frac{\partial L}{\partial b_{21}} = -2(3.68 - b_{21})$$

7.2 The sign decides the direction

Key point: At $b_{21} = -5$ the slope is -17.36 , so the update adds to b_{21} ; at $b_{21} = 5$ it is $+2.64$, so the update subtracts.

The algorithm cannot see the curve; it only knows the slope where it stands. Two cases:

- **Slope positive** (at $b_{21} = 5$, the slope is $+2.64$): raising b_{21} raises the loss. We must lower b_{21} , and $b_{21} - \eta \times 2.64$ does.
- **Slope negative** (at $b_{21} = -5$, the slope is -17.36): raising b_{21} lowers the loss. We must raise b_{21} , and $b_{21} - \eta \times (-17.36) = b_{21} + 17.36\eta$ does.

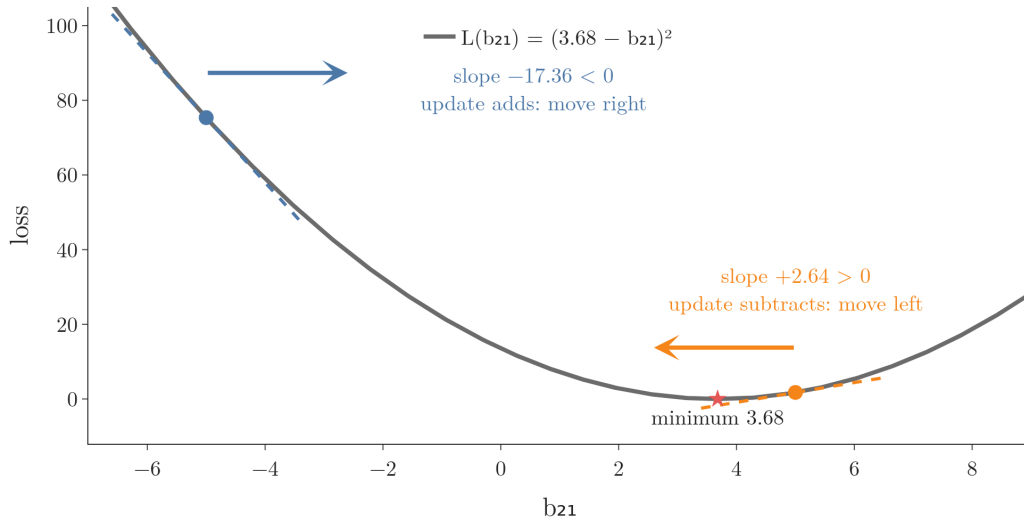


Figure 5: The same loss $L(b_{21})$ at two points: on the left the slope is negative and the update moves right; on the right it is positive and the update moves left

The minus sign does the right thing in both cases (Figure 5): we always move against the slope, in the direction of the **negative gradient** (the **gradient**, G-865, with its sign flipped). The step is also proportional to the slope's size: big where the curve is steep and far from the minimum, small near the bottom where it flattens.

8. Why we need a learning rate

Key point: Without a learning rate ($\eta = 1$), b_{21} jumps between -5 and 12.36 forever. $\eta = 0.1$ settles in; $\eta = 0.01$ crawls; $\eta = 1.1$ blows up.

Starting from $b_{21} = -5$, where the slope is -17.36 :

1. **In words:** the new value is the old value minus the learning rate times the slope.
2. **Formula:** $b_{21} \leftarrow b_{21} - \eta \partial L / \partial b_{21}$.
3. **Example:** with $\eta = 1$, $b_{21} = -5 - (-17.36) = 12.36$. There the slope is $-2(3.68 - 12.36) = 17.36$, so the next value is $12.36 - 17.36 = -5$, back at the start. With $\eta = 0.1$: $b_{21} = -5 + 0.1 \times 17.36 = -3.264$, a smaller and safer step.

Figure 1 runs 10 updates with four learning rates:

- $\eta = 0.01$: $-5, -4.83, -4.66, -4.49, \dots$, and -3.41 after 10 steps. Tiny steps: slow.
- $\eta = 0.1$: $-5, -3.26, -1.88, -0.76, \dots$, and 2.75 after 10 steps. The steps shrink as the slope flattens, and b_{21} converges to 3.68 .
- $\eta = 1$: $-5, 12.36, -5, 12.36, \dots$. The value zigzags across the bowl forever.
- $\eta = 1.1$: $-5, 14.10, -8.82, 18.68, \dots$, and -50.06 after 10 steps. Each jump overshoots more: the updates diverge.

Multiplying the slope by a small factor such as 0.1 or 0.01 turns the zigzag into small steps. As we near the minimum the slope shrinks, so the steps shrink with it. Too small a learning rate wastes time; too large overshoots and can run away. The learning rate (G-1068) is a **hyperparameter** (G-910), a setting we choose rather than learn, to tune (see section 5 of the [gradient descent Note](#)).

Extra: On this parabola the update is $b_{21} - 3.68 \leftarrow (1 - 2\eta)(b_{21} - 3.68)$: each step multiplies the distance to the minimum by $1 - 2\eta$. The factor is 0.8 for $\eta = 0.1$ (shrinks), -1 for $\eta = 1$ (same size, flipped side), and -1.2 for $\eta = 1.1$ (grows). Any η above 1 diverges here; how large a rate is safe depends on how steep the loss is.

In the real network all 9 parameters move at once, each against its own slope. The loss surface has 9 dimensions instead of one, but the reasoning is the same along every direction.

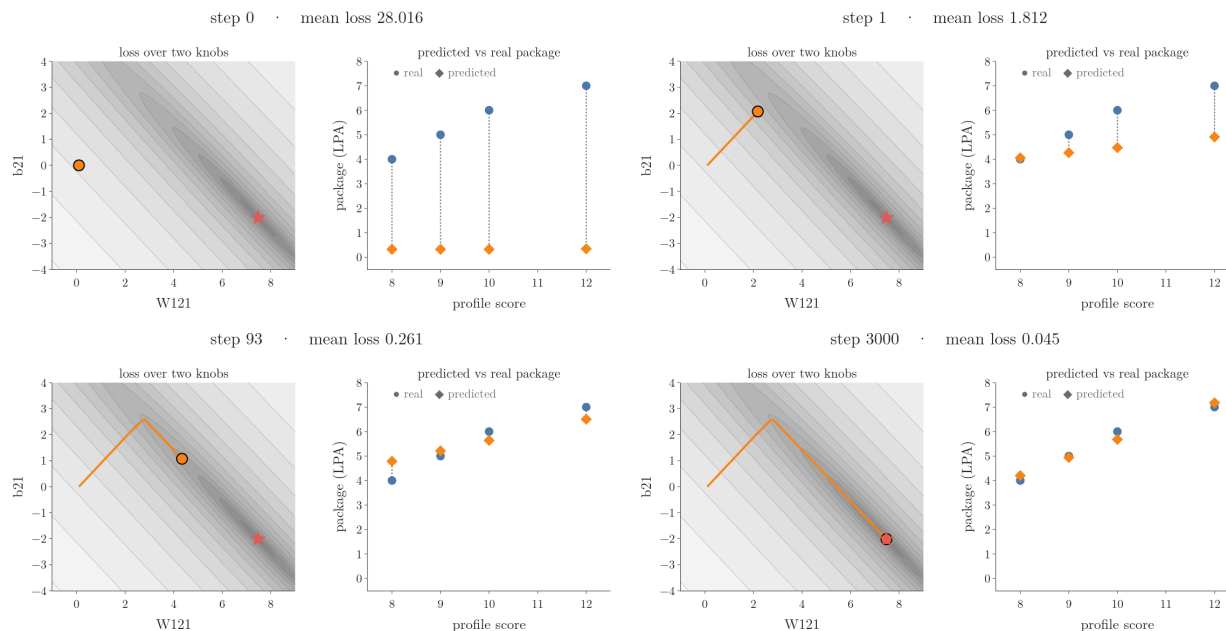


Figure 6: Gradient descent on two of the nine knobs at once, W_{21}^1 and b_{21} , over all four students; the other seven stay at their starting values. Left: the path on the contours of the mean loss (the star is the lowest point). Right: the predicted packages (diamonds) move onto the real ones (dots). Learning rate 0.2.

Figure 6 shows two parameters moving together, each against its own slope. Watch the first step: the ball drops straight into the long valley (mean loss 28.0 to 1.8) and the predictions jump up to the real packages. After that the slope along the valley is gentle, so the steps are small and the ball slides slowly towards the star: the mean loss is 0.25 after 100 steps and settles at 0.045 near step 1,000. The picture of a ball rolling down a loss surface follows Sanderson (3Blue1Brown), here redrawn with our own network.

9. When to stop: convergence

Key point: At the minimum the slope is zero, so $W_{\text{new}} = W_{\text{old}}$: updates stop changing anything. In practice we run a fixed number of epochs.

The algorithm has **converged** (**convergence**, G-472) when the updates stop changing the parameters: $W_{\text{new}} \approx W_{\text{old}}$. Convergence happens when $\eta \partial L / \partial W \approx 0$, that is, when the slope is close to zero: we are at a minimum.

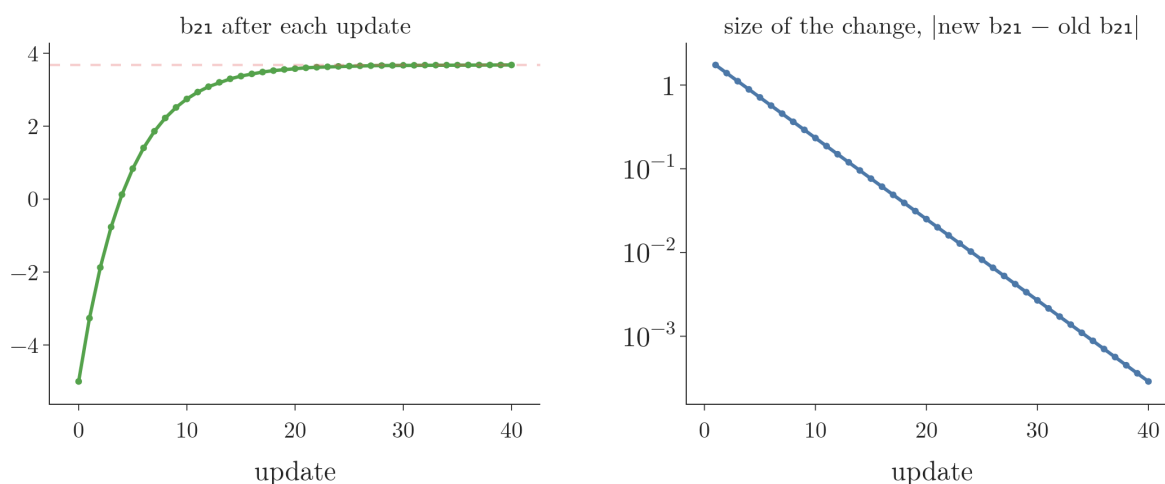


Figure 7: With learning rate 0.1: b_{21} after each update (left) and the size of each change (right, log scale)

Figure 7 shows convergence on $L(b_{21})$. Each change is 0.8 times the one before, so on the log scale the changes fall on a straight line, from 1.7 at the first update to 0.0003 at the 40th, while b_{21} flattens onto 3.68.

So the “right” loop is “repeat until convergence”. In practice we write a loop over a fixed number of epochs, such as 100 or 1,000, chosen large enough for the loss to settle (the stopping rules of section 2.3 of the [gradient descent Note](#)).

Extra: A slope of zero only proves a *local* minimum. A network’s loss is not one bowl, so gradient descent can settle in a dip that is not the lowest (see the [convex and non-convex cost functions Note](#)).

10. Summary

Question	Answer
What is the loss a function of?	All weights and biases; data and y are constants
What is the gradient?	The partial derivatives of the loss, one per parameter (9 here)
What does each derivative say?	How the loss changes with that parameter: size and sign
Why subtract it?	To move against the slope, downhill, whichever its sign
Why a learning rate?	To take small steps: $\eta = 1$ zigzags, $\eta = 1.1$ diverges, $\eta = 0.1$ converges
When to stop?	When updates no longer change the parameters; in practice after a fixed number of epochs

- Writing $\hat{y}$ out shows L as one function of the 9 parameters.
- We cannot solve “all derivatives = 0” for a network, so we walk downhill.
- The minus sign moves each parameter the right way; the slope’s size makes steps shrink near the minimum.
- On $L(b_{21}) = (3.68 - b_{21})^2$ each update multiplies the distance to the minimum by $1 - 2\eta$.

11. Sources

Built from

- CampusX, “Backpropagation Part 3 | The Why | Complete Deep Learning Playlist”, YouTube, <https://www.youtube.com/watch?v=6xO-x8y0YSY>
- Sanderson, G. (3Blue1Brown), “Gradient descent, how neural networks learn”, 3blue1brown.com/lessons/gradient-descent.

Other references

- Goodfellow, Bengio and Courville, *Deep Learning*, MIT Press, 2016, §6.2 (non-convex losses; iterative gradient-based training).

12. Key terms

Term	Meaning
Rate of change	How much one quantity changes per unit change of another; the meaning of a derivative
Negative gradient	The gradient with its sign flipped: the direction in which the loss falls fastest

Term	Meaning
Divergence	Updates that overshoot more and more, so the parameter and the loss run away
Convergence	The state where updates no longer change the parameters because the slope is about zero